IGNACIO JAVIER DIAZ PACCE

Product Engineer | IoT & MedTech | Hardware to Mobile

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PROFESSIONAL SUMMARY

Electronics engineer focused on end-to-end product development — from hardware and firmware to mobile applications. I build and scale systems that connect real devices with software, combining technical depth with business judgment. I work autonomously, take full ownership of outcomes, and use AI tools to increase delivery speed and quality.

WORK EXPERIENCE

Product Engineer — Firmware, Software & Development

Nov 2024 – Present

Electromedicina Morales S.R.L.

Buenos Aires

Part of the design and development team for veterinary biomedical devices, covering firmware, mobile software, and production optimization.

Software Development

- Led the design and development of EVET Hub, a Flutter app (iOS/Android) with a hub architecture to connect several devices; the first one integrated is a veterinary electrocardiograph.
- Built a PWA for quoting and lead capture at trade shows (React, TypeScript, Supabase), with QR code per product, ROI per event, offline mode and sync with the company ERP SQL database.
- Built a desktop app (Python, PyQt5, SQLite) to flash firmware through ST-Link/SWD and SPI memories, with batch flashing, role login and a Windows installer.
- Implemented production traceability: every flash is recorded and synced on the network, and linked with the ERP SQL database to know which board has firmware and which customer received the device.

Firmware

- STM32 firmware for a veterinary electrocardiograph. Migration from 1 to 3 channels for greater diagnostic capability.
- Corrective and evolutionary maintenance of existing firmware for the product catalog.

Process & Production Optimization

- Automated technical documentation and led the outsourcing of PCBA manufacturing
- Reactivated project management tools (Redmine)
- Procurement management for the development area
- Implemented document organization system
- Set up the design file structure and defined the requirements for a new magnetotherapy device

Design, Development & Production Manager

Apr 2022 – Oct 2024

StandFit Equipment

Goya, Corrientes

Built from scratch the operational infrastructure of a fitness equipment company: product design, production processes, inventory, and control systems.

- Managed a portfolio of 30+ fitness equipment models, with original design and redesign based on customer feedback (Voice of Customer).
- Designed a 6-station production layout. Tripled production capacity from 1–2 to 6 machines per month with the same headcount.
- Led and trained a team of 4 people with no prior experience, progressively delegating daily coordination to an operational team lead.
- Implemented digital inventory with barcode (100+ SKUs) and developed from scratch production orders, BOM costing, purchase planning and preventive maintenance.
- Reduced component costs by replacing machining with additive manufacturing (3D printing).

PCB Design Intern (Professional Practice) Dec 2021 – Mar 2022

Voltu Motor Inc. Paraná, Entre Ríos

- Design of electronic boards for BMS testing in electric mobility. Electrical validation and 3D modeling.

PROJECTS & COLLABORATIONS

NutriBalanz nutribalanz.com.ar

- Platform for nutritionists, in real use: professional and patient views, appointments, and push and email reminders. React, TypeScript, Supabase (PostgreSQL with RLS), PWA.

Nutrilina Tips nutrilinatips.com.ar

- Website for a nutritionist built with Astro, optimized for SEO and AEO: JSON-LD structured data, sitemap and llms.txt.

Novha Clothing store

- Astro website with local SEO and a multi-business point-of-sale system (sales, cash register, stock, customer accounts, offline mode).

FaunX App Store · Google Play

- Partner and member of the development team of a digital pet passport app (App Store and Google Play); I contribute to the product vision and I am starting to join the development.

EDUCATION

Electronics Engineer 2016 – 2023

Universidad Nacional del Nordeste (UNNE) Corrientes

- GER Scholar: control of grid-connected photovoltaic systems (2021–2022). Teaching Assistant in Electrotechnics II (2021).

Additional: Technical English (UNNE) · Public Speaking (UNNE) · Project Management & SysML (in progress, Udemy)

TECHNICAL SKILLS

Firmware & Embedded	C (STM32 ARM Cortex-M, ATmega328P), FreeRTOS, UART, SPI, I2C, BLE, STM32CubeProgrammer (ST-Link/SWD)
Mobile & Web Dev	Flutter/Dart, Android Studio, React, TypeScript, Astro, Vite, Tailwind CSS, PWA, SEO/AEO
Electronics Design	KiCad (advanced), Proteus, LTspice, MultiSim
3D Modeling	Fusion 360, SolidWorks, Cura, OrcaSlicer, 3D printing
Automation	Python (PyQt5), Node.js, PowerShell, SQL Server, PostgreSQL, SQLite, Google Apps Script, Google Sheets API
AI & Dev Workflow	Claude Code (CLI), Cursor, MCP Servers (Vercel · GitHub · Supabase), Vercel, Supabase
Tools	Git, Bitbucket, GitHub, Redmine, Matlab, Simulink, Google Workspace, MS Office

MANAGEMENT SKILLS

Team leadership · Cross-disciplinary coordination · Stakeholder management · Continuous improvement · Production planning · Change management · Technical documentation · Production traceability · Supply chain · BOM costing · Ownership

LANGUAGES & ADDITIONAL INFO

Spanish Native **English** Basic (B1 – in active learning) **Availability** Remote · Hybrid Buenos Aires

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